ARULRAJ VENKADACHALAM

SENIOR FRONTEND ENGINEER | FRONTEND ARCHITECT

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PROFESSIONAL SUMMARY

Senior Frontend Engineer with 6+ years designing and delivering scalable web and mobile platforms, leading frontend architecture across 16 frontend applications and modules at The Reward Store. Architected a Module Federation platform of 9 micro-frontends integrated into 4 host applications, a multi-gateway checkout and a 76-component multi-brand design system. Owns release engineering, testing standards and production reliability.

KEY STRENGTHS

- Frontend Architecture & Micro-Frontends
- Design Systems & Multi-brand Theming
- Release & Production Ownership
- Mobile Apps & SDKs
- Payment & Checkout Flows
- Testing & Accessibility
- Performance & Bundle Budgets
- Technical Leadership & Mentoring

PROFESSIONAL EXPERIENCE

SENIOR SOFTWARE ENGINEER, FRONTEND

JAN 2025 – PRESENT

The Reward Store, Global Rewards Platform (Vananam Rewards Private Limited)

Leads frontend architecture for three products, Storefront (Rewards), Rekyndl (Loyalty) and Applaud IQ (Employee Recognition), plus the Client Panel and Admin Panel: 16 frontend applications and modules (9 micro-frontends, 4 hosts, 3 standalone apps). Built the Rekyndl and Applaud IQ apps from scratch; started as the only senior frontend engineer and now also mentors a junior developer.

- **Micro-frontend architecture:** Built the 9 Storefront micro-frontends in React and TypeScript on Rspack Module Federation and composed them into the 4 hosts (Storefront, Rekyndl, Applaud IQ, Client Panel) with shared singleton dependencies, so features are built once and reused.
- **Payments & checkout:** Built one checkout for six categories (gift cards, recharge, bills, flights, merchandise, experiences) with a per-customer payment gateway resolver (points, wallet, Razorpay, Cashfree, Zaakpay), status polling, failure / dismissal / refund handling and a shared error catalog.
- **Release isolation & performance:** Made each remote independently releasable with per-remote Amplify rollback, federation retry, route-level error boundaries, stale-chunk recovery and release detection; targeted splitChunks and Brotli bundle budgets fail CI when a build exceeds its budget.
- **Design system:** Designed and published @therewardstore/*, used by all 16 apps and modules: 76 accessible Radix + Tailwind v4 components with multi-brand light/dark theming and runtime white-labelling; 80 versions in 11 weeks, replacing two legacy UI kits.
- **Testing & accessibility:** Established ~460 Playwright tests with @axe-core accessibility checks against Storybook and a 95% changed-file coverage gate on Vitest + React Testing Library suites, with CI gates that block detected test failures and accessibility violations.
- **Mobile & SDKs:** Published the Storefront and Applaud IQ apps on a remote-URL Capacitor shell, so web deploys reach installed apps without a store review cycle; shipped the Applaud IQ embed SDK for 6 runtimes on one bridge protocol.
- **Production & rollout:** Owns AWS Amplify deployments, staging / production and store releases; per-customer feature flags gate which remotes load and stage rollouts. Diagnosed a Capacitor session-cookie loss through HyperDX / ClickHouse traces and fixed it with a dev API proxy.
- **Architecture leadership:** Owns the frontend platform decisions all 16 apps and modules build on (Module Federation conventions, shared packages), recorded in 45+ architecture decision records; sets coding standards, reviews PRs and built the create-app scaffolding CLI.
- **Backend contributions (AI-assisted):** Over ~6 months, contributed Go / gRPC routes, protobuf contracts, handlers and PostgreSQL migrations to 5 services (gateway, notifications, employee, rewards, loyalty-engine) and Kafka consumers; reviewed and tested every change.

SOFTWARE ENGINEER JUL 2023 – JAN 2025

Ripple Street India Private Limited, B2H Marketing Platform

React web platform on Module Federation micro-frontends with AWS Amplify and AppSync GraphQL, plus iOS and Android React Native apps.

- **Apollo Client & GraphQL:** Kept lists current without refetching using cache.modify, keyFields and optimisticResponse (rollback with an error toast on failure); regenerated TypeScript types from the AppSync schema with Amplify codegen.
- Shipped the iOS and Android React Native apps to both stores.
- Owned core features and removed redundant code, cutting bundle size by 20%.

SENIOR SOFTWARE ENGINEER APR 2022 – JUL 2023

Squash Apps, FinTech & Workplace Solutions

- Led a team of 2 engineers delivering the Payfiniti payment apps for web, Android and iOS on time.
- Built an offline-capable office desk-booking app for Android and iOS with React Native CLI.
- Independently managed two client projects, one for a US client and one for an Australian client, owning requirements discussions, development coordination and delivery across time zones.

FRONTEND DEVELOPER OCT 2021 – MAR 2022

Zita.ai, AI-Powered Recruitment Platform

- Owned the entire frontend of an AI resume-evaluation platform in React.js, from UI/UX to production.
- Streamlined the evaluation flow, cutting candidate selection time by 50% versus the manual process.

FRONTEND DEVELOPER MAY 2020 – OCT 2021

ShopUp, Retailer E-Commerce Platform

- Built a UI component npm library standardising UI across projects (+30% developer productivity).
- Designed and implemented the product-management interface for the retailer catalogue.

TECHNICAL SKILLS

- **Frontend:** React 19, TypeScript, JavaScript, Next.js, Vite, React Router, Redux Toolkit / RTK Query, Apollo Client, Tailwind CSS v4, Radix UI, SCSS / Sass, Styled Components, CSS Modules, Formik + Yup, React Hook Form + Zod
- **Architecture & Build:** Micro-frontends, Module Federation, Rspack / Webpack (splitChunks, code splitting, Brotli bundle budgets), design systems, pnpm / Turborepo monorepos, Changesets
- **Payments & APIs:** Razorpay, Cashfree, Zaakpay, points and wallet checkout; REST, OpenAPI (generated types), GraphQL (Apollo, AWS AppSync), SSE, OAuth / SSO / OTP
- **Testing & Accessibility:** Playwright, @axe-core/playwright, Vitest, Jest, React Testing Library, Storybook, CI coverage gates
- **Mobile, Cloud & Delivery:** Capacitor, React Native, App Store / Google Play releases, SDKs for iOS / Android / Flutter; AWS Amplify, GitLab CI, Bitbucket Pipelines, feature flags, HyperDX / ClickHouse
- **Backend & Tools:** Go / gRPC, PostgreSQL, Kafka (working knowledge); Claude Code, Cursor

EDUCATION

B.E. — Computer Science Engineering 2017 – 2020

Thiagarajar College of Engineering, Madurai · CGPA: 7.03

Diploma — Electronics & Communication Engineering 2015 – 2017

Thiagarajar Polytechnic, Salem · Percentage: 89%

LANGUAGES

Tamil (Native) | English (Professional Working Proficiency)